



Copper-catalyzed amination of potassium aryl trifluoroborates using aqueous ammonia

André P. Liesen, Arisson T. Silva, jokderléa C. Sousa, Paulo H. Menezes, Roberta A. Oliveira*

Departamento de Química Fundamental, Universidade Federal de Pernambuco, Recife, PE 50740-540, Brazil

ARTICLE INFO

Article history:

Received 3 May 2012

Revised 30 May 2012

Accepted 4 June 2012

Available online 12 June 2012

Keywords:

Potassium organotrifluoroborates

Aromatic amines

Aqueous ammonia

ABSTRACT

The conversion of potassium aryl trifluoroborates containing different functionalities into the corresponding aryl amines using a catalytic amount of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is described. The methodology uses water as a solvent under aerobic conditions to give the products in good yields.

© 2012 Elsevier Ltd. All rights reserved.

Aryl amines are important building blocks in pharmaceutical, agrochemical, and polymer industries.¹ The most common approach to prepare this class of compounds generally relies on the use of transition-metal catalysts to promote the coupling reaction of aryl halides with ammonia or ammonia derivatives.²

Several other methods based on the metal-catalyzed reduction of nitro aryl³ or aryl azides⁴ by hydrogenation, electrochemical,⁵ and hydride transfer conditions⁶ were also described. For economical and practical reasons, the use of ammonia is the most convenient and efficient approach. However, the synthesis of aniline derivatives using ammonia has several drawbacks such as the ligand displacement from the catalyst by ammonia to form catalytically unreactive species,⁷ harsh reaction conditions, and the formation of undesirable diaryl and triaryl amines by-products.⁸

Although the aryl halides are still the main substrates to perform the reaction, boronic acids are also particularly attractive synthetic intermediates⁹ due to their unique properties as mild organic Lewis acids and lower reactivity. However, the tricoordinate nature of these compounds makes them moisture sensitive, limiting its application in multistep sequences. In addition, boronic acids possess some limitations such as the difficulty to purify and some uncertainty about stoichiometry, due to the formation of boroximes.¹⁰

Organotrifluoroborates have proven to be a good option to replace boronic acids and boronate esters in transmetallation reactions, providing many advantages over the latter reagents.¹¹ They are very useful reagents for C–C bond formation¹² and there are

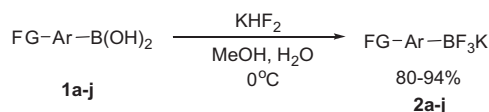
many possibilities of functional group interconversion reactions involving these compounds including the syntheses of aldehydes,¹³ ketones,¹⁴ alcohols,¹⁵ alkynes,¹⁶ and alkenes,¹⁷ all containing the organotrifluoroborate moiety, which can be later used in Suzuki–Miyaura type couplings.

The complete characterization of these salts, including heteronuclei NMR analysis, has proven to be possible since better NMR pulse sequences were developed, allowing the acquisition of new data related to ^{19}F – ^{11}B coupling constants.¹⁸ In addition, exact mass measurements were obtained for a variety of potassium organotrifluoroborates using a sector ESI mass spectrometer.¹⁹

It is also important to emphasize the recent interest on biological activities and toxicological data related to these salts. It is already described that organotrifluoroborates may act as serine protease inhibitors,²⁰ growth inhibitors on human tumor cell lines,²¹ and some of them appear to have antinociceptive activity.²²

Herein, we wish to describe an environmentally benign reaction for the synthesis of aryl amines based on the reaction of potassium aryl trifluoroborates and aqueous ammonia under aerobic conditions catalyzed by copper (II).

The potassium organotrifluoroborates **2a–j** used in this study were prepared from the corresponding boronic acids **1a–j** using



Scheme 1.

* Corresponding author. Tel.: +55 81 2126 7473; fax: +55 81 2126 8442.

E-mail address: roberta.aoliveira@ufpe.br (R.A. Oliveira).

KHF₂ in MeOH:H₂O in good yields²³ and their structures were confirmed by ¹H, ¹³C, ¹¹B, and ¹⁹F NMR analysis (Scheme 1).

Our initial studies focused on the development of an optimum set of reaction under green chemistry conditions. For this purpose, potassium 3-nitro-phenyltrifluoroborate (**2e**) (0.25 mmol) was treated with different catalysts at 25 °C for 24 h and aqueous ammonia (4 mL of a 28–30% solution) was used as amino group source with sodium hydroxide (3 equiv) as base. The obtained product **3e** was characterized by ¹H and ¹³C NMR. The results are described in Table 1.

As shown in Table 1, the use of copper (II) salts gave the corresponding product in moderate to good yields (Table 1, entries 2–7). The best results were obtained when CuSO₄·5H₂O was used in concentrations of 10 and 20 mol %, **3e** being obtained in high yields in both cases. When Cu(OAc)₂ was used as a catalyst lower yields were observed (Table 1, entries 5–7).

When the reaction was performed with Pd(OAc)₂ no product was observed in the applied concentrations, probably due to the formation of catalytically unreactive species with ammonia. All reactions were monitored by ¹H NMR until the disappearance of the starting material which was accomplished after 24 h.

In this way, the suitable catalyst for the proposed reaction was CuSO₄·5H₂O in a concentration of 10 mol %. By extending the reaction conditions to other potassium aryl trifluoroborates, various functionalized aryl amines were obtained in moderate to good yields (Table 2) and the formation of the products was confirmed by the analysis of the ¹H and ¹³C NMR spectra.

Inspection of Table 2 shows that the method is tolerant to a wide range of functional groups, being more efficient when electron-withdrawing and neutral groups were present in the starting potassium aryl trifluoroborate. Thus, substrates with the ring containing halogens and cyano, nitro, acetyl, and formyl groups gave the corresponding functionalized aryl amines in good to moderate yields (Table 2, entries 1–6).

The applications of naphthylamines in the dyestuff industry²⁴ and in heterocyclic chemistry as building blocks²⁵ are well known and the development of efficient synthetic methods to prepare this class of compounds is of great interest. In this context, compound **3h** was prepared in good yield and high purity using the reaction conditions (Table 2, entry 8).

When a sterically hindered di-*ortho*-substituted aryl trifluoroborate **2i** was used, the corresponding aryl amine was obtained in moderate yield proving, in this case, that the reaction is not only sensitive to the electronic, but also to steric effects (Table 2, entry 9).

When the electron-donating potassium aryl trifluoroborate **2j** was used, the corresponding product **3j** was not observed (Table 2, entry 10).

Table 1
Amination of potassium aryl trifluoroborates using aqueous ammonia: influence of catalyst

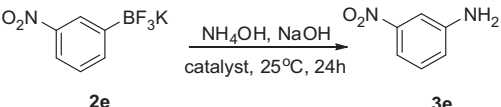
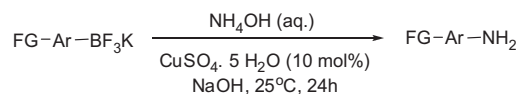
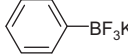
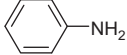
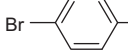
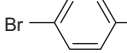
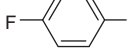
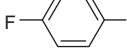
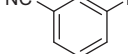
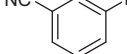
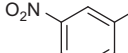
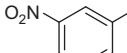
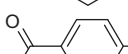
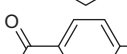
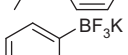
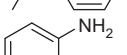
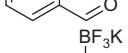
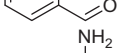
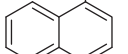
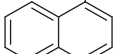
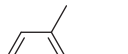
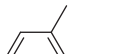
		
Entry	Catalyst (mol %)	3e (%)
1	—	0
2	CuSO ₄ ·5H ₂ O (5)	50
3	CuSO ₄ ·5H ₂ O (10)	90
4	CuSO ₄ ·5H ₂ O (20)	90
5	Cu(OAc) ₂ (5)	50
6	Cu(OAc) ₂ (10)	60
7	Cu(OAc) ₂ (20)	65
8	Pd(OAc) ₂ (5)	0
9	Pd(OAc) ₂ (10)	0
10	Pd(OAc) ₂ (20)	0

Table 2

Aryl amines (**3**) prepared from the copper-catalytic amination of ArBF₃K (**2**)



Entry	FG-Ar-BF ₃ K	FG-Ar-NH ₂	Yield ^a (%)
1	2a 	3a 	75
2	2b 	3b 	80
3	2c 	3c 	85
4	2d 	3d 	88
5	2e 	3e 	90
6	2f 	3f 	92
7	2g 	3g 	50
8	2h 	3h 	83
9	2i 	3i 	40
10	2j 	3j 	— ^b

^a Isolated yield.

^b The corresponding product was not observed.

Recently, the kinetics of hydrolysis of RBF₃K to the corresponding boronic acids in the presence and absence of base, buffers, and glass were studied in detail by Lloyd-Jones and co-workers in the context of their application in Suzuki–Miyaura coupling.²⁶ They observed that electron-rich aryl trifluoroborates undergo fast or very fast hydrolysis, while electron-poor aryl trifluoroborates are hydrolyzed very slowly.

In our case, this observation together with the fact that electron-rich aryl boronic acids can undergo a protodeboration reaction via a S_E2 mechanism, the proton transfer being the rate-determining step followed by the rapid ionic cleavage of C–B bond,²⁷ that could explain the results observed in Table 2, where better yields were observed when electron-withdrawing groups were present in the aromatic ring.

In summary, we have demonstrated that the potassium aryl trifluoroborates are useful substrates for the synthesis of the corresponding aryl amines in good yields under mild conditions. The green methodology is simple,²⁸ uses water as solvent, low catalyst loading, and is synthetically useful while it could be applied for the synthesis of more complex aromatic amines.

Acknowledgments

The authors gratefully acknowledge FACEPE (APQ-1402-1.06/10), CNPq (484778/2011-0), CAPES, and INCT-INAMI for financial

support. P.H.M and A.P.L. are thankful to CNPq for their fellowships.

Supplementary data

Supplementary data (experimental procedures and spectroscopic characterization data, as well as ^1H , ^{13}C , ^{19}F and ^{11}B NMR spectra for all synthesized compounds) associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.tetlet.2012.06.011>. These data include MOL files and InChIKeys of the most important compounds described in this article.

References and notes

- (a) Schlummer, B.; Scholz, U. *Adv. Synth. Catal.* **2004**, *346*, 1599–1626; (b) Meng, F.; Zhu, X.; Li, Y.; Xie, J.; Wang, B.; Yao, J.; Wan, Y. *Eur. J. Org. Chem.* **2010**, 6149–6152.
- Aubin, Y.; Fischmeister, C.; Thomas, C. M.; Renaud, J. L. *Chem. Soc. Rev.* **2010**, *39*, 4130–4145.
- (a) Pehlivan, L.; Métay, E.; Laval, S.; Dayoub, W.; Demonchaux, P.; Mignani, G.; Lemaire, M. *Tetrahedron Lett.* **2010**, *51*, 1939–1941; (b) Lee, H.-Y.; An, M. *Bull. Korean Chem. Soc.* **2004**, *25*, 1717–1719; (c) Khan, F. A.; Sudheer, C. *Tetrahedron Lett.* **2009**, *50*, 3394–3396; (d) Zhou, Y.; Li, J.; Liu, H.; Zhao, L.; Jiang, H. *Tetrahedron Lett.* **2006**, *47*, 8511–8514; (e) Gamble, A. B.; Garner, J.; Gordon, C. P.; O'Connor, S. M. J.; Keller, P. A. *Synth. Commun.* **2007**, *37*, 2777–2786.
- Norcliffe, J. L.; Conway, L. P.; Hodgson, D. R. W. *Tetrahedron Lett.* **2011**, *52*, 2730–2732.
- Kokkinidis, G.; Papoutsis, A.; Papanastasiou, G. J. *Electroanal. Chem.* **1993**, *359*, 253–271.
- (a) Ben Taleb, A.; Jenner, G. J. *Mol. Catal.* **1994**, *91*, L149–L153; (b) Sharma, U.; Kumar, P.; Kumar, N.; Kumar, V.; Singh, B. *Adv. Synth. Catal.* **2010**, *352*, 1834–1840; (c) Watanabe, Y.; Ohta, T.; Tsuji, Y.; Hiyoshi, T.; Tsuji, Y. *Bull. Chem. Soc. Jpn.* **1984**, *57*, 2440–2444; For the asymmetric version of the reaction see: (d) Gladiali, S.; Alberico, E. *Chem. Soc. Rev.* **2006**, *35*, 226–236.
- (a) Widenhoefer, R. A.; Buchwald, S. L. *Organometallics* **1996**, *15*, 2755–2763; (b) Widenhoefer, R. A.; Buchwald, S. L. *Organometallics* **1996**, *15*, 3534–3542; (c) Paul, F.; Patt, J.; Hartwig, J. F. *Organometallics* **1995**, *14*, 3030–3039.
- Shen, Q.; Hartwig, J. F. *J. Am. Chem. Soc.* **2006**, *128*, 10028–10029.
- (a) Cazorla, C.; Métay, E.; Lemaire, M. *Tetrahedron* **2011**, *67*, 8615–8621; (b) Yang, H.; Li, Y.; Jiang, M.; Wang, J.; Fu, H. *Chem. Eur. J.* **2011**, *17*, 5652–5660; (c) Jiang, Z.; Wu, Z.; Wang, L.; Wu, D.; Zhou, X. *Can. J. Chem.* **2010**, *88*, 964–968; (d) Rao, H.; Fu, H.; Jiang, Y.; Zhao, Y. *Angew. Chem., Int. Ed.* **2009**, *48*, 1114–1116; (e) Quach, T. D.; Batey, R. A. *Org. Lett.* **2003**, *5*, 4397–4400; (f) Matteson, D. S.; Kim, G. Y. *Org. Lett.* **2002**, *4*, 2153–2155.
- Molander, G. A.; Ellis, N. *Acc. Chem. Res.* **2007**, *40*, 275–286.
- (a) Molander, G. A.; Figueroa, R. *Aldrichim. Acta* **2005**, *38*, 49–56; (b) Stefani, H. A.; Cella, R.; Vieira, A. S. *Tetrahedron* **2007**, *63*, 3623–3658; (c) Darses, S.; Genet, J.-P. *Chem. Rev.* **2008**, *108*, 288; (e) Oliveira, R. A. *Synlett* **2009**, 505–506.
- Santos-Filho, E. F.; Sousa, J. C.; Bezerra, N. M. M.; Menezes, P. H.; Oliveira, R. A. *Tetrahedron Lett.* **2011**, *52*, 5288–5291.
- Molander, G. A.; Petrillo, D. E. *J. Am. Chem. Soc.* **2006**, *128*, 9634–9635.
- Molander, G. A.; Cooper, D. J. *J. Org. Chem.* **2007**, *72*, 3558–3560.
- Molander, G. A.; Figueroa, R. *Org. Lett.* **2006**, *8*, 75–78.
- Katona, B. W.; Machrouhi, F.; Molander, G. A. *J. Org. Chem.* **2002**, *67*, 8416–8423.
- (a) Molander, G. A.; Oliveira, R. A. *Tetrahedron Lett.* **2008**, *49*, 1266–1268; (b) Molander, G. A.; Figueroa, R. *J. Org. Chem.* **2006**, *71*, 6135–6140; (c) Molander, G. A.; Ham, J.; Canturk, B. *Org. Lett.* **2007**, *9*, 821–824.
- Oliveira, R. A.; Silva, R. O.; Molander, G. A.; Menezes, P. H. *Magn. Res. Chem.* **2009**, *47*, 873–878.
- Petrillo, D. E.; Kohli, R. K.; Molander, G. A. *J. Am. Soc. Mass Spectrom.* **2007**, *18*, 404–405.
- Smoum, R.; Rubinstein, A.; Srebnik, M. *Org. Biomol. Chem.* **2005**, *3*, 941–944.
- Queiroz, M.-J. R. P.; Calhelha, R. C.; Vale-Silva, L. A.; Pinto, E.; Lima, R. T.; Vasconcelos, M. H. *Eur. J. Med. Chem.* **2010**, *45*, 5628–5634.
- Oliveira, R. A.; Savegnago, L.; Jesse, C. R.; Menezes, P. H.; Molander, G. A.; Nogueira, C. W. *Basic Clin. Pharmacol. Toxicol.* **2009**, *104*, 448–454.
- Vedejs, E.; Fields, S. C.; Hayashi, R.; Hithcock, S. R.; Powell, D. R.; Schrimpf, M. R. *J. Am. Chem. Soc.* **1999**, *121*, 2460–2470.
- Guinot, S. G. R.; Hepworth, J. D.; Wainwright, M. *Dyes and Pigments* **1998**, *36*, 387–393.
- Yépez, A. F.; Palma, A.; Stashenko, E.; Bahsasb, A.; Amaro-Luis, J. M. *Tetrahedron Lett.* **2006**, *47*, 5825–5828.
- Lennox, A. J. J.; Lloyd-Jones, G. C. *J. Am. Chem. Soc.* **2012**, *134*, 7431–7441.
- (a) Kuivila, H. G.; Nahabedian, K. V. *J. Am. Chem. Soc.* **1961**, *83*, 2159–2163; (b) Kuivila, H. G.; Nahabedian, K. V. *J. Am. Chem. Soc.* **1961**, *83*, 2164–2166; (c) Nahabedian, K. V.; Kuivila, H. G. *J. Am. Chem. Soc.* **1961**, *83*, 2167–2174; (d) Klingensmith, L. M.; Bio, M. M.; Moniz, G. A. *Tetrahedron Lett.* **2007**, *48*, 8242–8245.
- Representative procedure for the synthesis of 3a:** In a flask, at room temperature and under aerobic conditions, containing a mixture of potassium aryltrifluoroborate **2a** (1 mmol, 184 mg), $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (10 mol %, 25 mg) and NaOH (3 mmol, 120 mg) was added NH_4OH (4.0 mL of a 28–30% solution). The contents were stirred for 24 h. After this period, the reaction was extracted with dichloromethane (3×10 mL). The combined organic layers were washed with 1 M HCl (30 mL) and to the aqueous layer was added solid NaHCO_3 (until pH ≈ 9). The resulting mixture was again extracted with dichloromethane (3×10 mL) and the combined organic layers were dried over anhydrous MgSO_4 and filtered. The solvent was removed *in vacuo* to yield 69 mg of **3a** (75%) sufficiently pure for characterization. ^1H NMR (300 MHz; CDCl_3) δ 7.34 (t, $J = 7.7$ Hz, 2H), 6.95 (t, $J = 7.7$ Hz, 1H), 6.79 (d, $J = 8.5$ Hz, 2H), 3.66 (s, 2H); ^{13}C NMR (75 MHz, CDCl_3) δ 146.2, 129.0, 118.0, 114.8.